Yes! The block diagram is functionally complete and correctly structured for a single USB Type-C port supporting charging, high-speed/low-speed data transfer, and analog audio. Here is a detailed verification of your architecture, followed by what you need to consider before implementing it in OrCAD Capture.

Architectural Verification

* Power & Charging Path (VBUS & GND)
  * Component: MAX77751
  * Verification: Correct. Connecting VBUS and GND directly to the MAX77751 handles battery charging. As noted previously, the MAX77751 includes built-in over-voltage protection (OVP), eliminating the need for a separate external OVP IC.
* Configuration & SuperSpeed Data Path (CC1/CC2 & High-Speed Lanes)
  * Components: TUSB322 (CC Controller) + HD3SS3212 (SuperSpeed Mux)
  * Verification: Correct. The TUSB322 manages cable attach/detach detection, role configuration, and orientation detection. Its orientation output properly controls the HD3SS3212 2:1 differential switch, ensuring that the SuperSpeed TX/RX differential pairs route correctly to the Main Processor USB 3.0 controller regardless of how the reversible Type-C cable is plugged in.
* USB 2.0 & Analog Audio Multiplexing (D+/D-)
  * Component: TS5USBA224 (Analog Switch/MUX)
  * Verification: Correct. The TS5USBA224 is ideally suited for this application. It allows the D+/D- pins from the USB-C connector to dynamically multiplex between the Main Processor USB 2.0 transceiver and an Audio Codec/DAC (for analog audio output/headset accessory mode).

Is it Sufficient for OrCAD Capture Circuit Design?

Yes, this is a solid foundational architecture to start your OrCAD Capture schematic. However, to make it production-ready and fully robust for PCB layout in Cadence Allegro, ensure you add the following necessary blocks and nets during schematic entry:

* ESD Protection (TVS Diodes): Add robust ESD protection arrays on the VBUS, CC1, CC2, D+, D-, and SuperSpeed lines close to the USB-C connector receptacle to protect your ICs from electrostatic discharge.
* CC Pin Pull-Downs/Pull-Ups: Ensure the TUSB322 configuration pins (such as ID, EN, or mode-setting pins like AUDIO_EN / configuration strapping resistors) are properly tied off or configured according to whether the port is configured as DFP, URP, or Dual-Role

Port (DRP).

* Control / I2C Lines: Check if your TUSB322 or MAX77751 requires I2C configuration or interrupt (INT) / alert lines routed to your main processor for status monitoring and fault handling.

* Decoupling Capacitors: Place appropriate high-frequency decoupling ceramic capacitors (e.g., 0.1\,\mu\text{F} and 1\,\mu\text{F}) on the power pins of the TUSB322, HD3SS3212, and TS5USBA224.

* AC Coupling Capacitors: Ensure proper AC-coupling capacitors (typically 100\,\text{nF}) are placed on the SuperSpeed TX/RX differential lines between the connector/mux and the processor as required by the USB 3.0 specification.

To successfully implement your architecture in OrCAD Capture, here is the precise pin-to-pin connection guide for each IC based on your system requirements (Charging, High-Speed Data, Audio, and CC Control).

1. USB Type-C Receptacle (The Central Node)

Make sure you use a full-featured 24-pin Type-C receptacle symbol in OrCAD.

* VBUS: Connect to MAX77751 (CHGIN / VBUS input pin) and input bulk/decoupling capacitors.

* GND: Connect to system Ground (PGND/GND plane).

* CC1 & CC2: Connect in parallel to TUSB322 (CC1 and CC2 pins) and MAX77751 CC pins (if using the variant with integrated CC detection).

* TX1+, TX1-, RX1+, RX1-, TX2+, TX2-, RX2+, RX2-: Connect to the connector-side pins of the HD3SS3212 SuperSpeed Mux.

* D+ & D-: Connect to the common port pins of the TS5USBA224 analog switch.

2. TUSB322 (CC Controller / Plug Orientation Detector)

Purpose: Detects cable orientation and configures DFP/UFP/DRP mode.

* CC1 / CC2: Connect directly to CC1 and CC2 lines from the Type-C Receptacle.

* VDD: Connect to stable 3.3V power rail. Place a 0.1\,\mu\text{F} decoupling capacitor close to the pin.

* GND: Connect to Ground.

* DIR (Direction Status Output): Connect to the orientation control pin of the

HD3SS3212 mux (SEL pin) to automatically steer high-speed lanes based on how the cable is flipped.

* AUDIO_EN / Config Pins: Pull-up or pull-down depending on whether you want Accessory Audio Mode enabled (tie appropriately per datasheet truth table to enable audio adapter routing).

3. HD3SS3212 (SuperSpeed USB 3.0 Mux/Demux)

Purpose: 2:1 differential switch routing high-speed USB 3.0 lines dynamically based on cable flip.

* VCC: Connect to 3.3V rail with a 0.1\,\mu\text{F} decoupling capacitor.

* GND / EP (Exposed Pad): Connect firmly to the ground plane.

* Connector Side Ports (C0p/C0n, C1p/C1n): Connect to the differential SuperSpeed TX/RX pairs coming from the USB-C receptacle.

* Processor Side Ports (B0p/B0n, B1p/B1n or A ports): Connect through 100\,\text{nF} AC-coupling capacitors to the Main Processor USB 3.0 TX/RX pins.

* SEL (Port/Orientation Select): Connect directly to the DIR output pin of the TUSB322 controller.

* OEn (Output Enable): Tie to GND (Active-Low) to keep the switch permanently enabled during operation.

4. TS5USBA224 (USB 2.0 & Analog Audio Mux)

Purpose: Multiplexes D+/D- lines between the USB 2.0 processor interface and the Audio Codec/DAC.

* VCC: Connect to 3.3V supply rail (with local decoupling capacitor).

* GND: Connect to Ground.

* Common D+ / D- Pins: Connect to the D+ and D- data lines from the USB-C receptacle.

* Path 1 (USB 2.0 Data): Connect to the Main Processor USB 2.0 transceiver pins (USB_D+, USB_D-).

* Path 2 (Audio): Connect to the Audio Codec / DAC analog outputs/inputs (HP_R, HP_L or differential audio lines).

* Control Pins (SEL / OE): Driven by a GPIO from your main processor or the audio detection logic to switch between data mode and audio accessory mode.

5. MAX77751 (Standalone USB-C Charger & Power Path)

Purpose: Manages 5V VBUS power delivery, internal OVP, and single-cell Li-Ion battery

charging.

* CHGIN / VBUS: Connect directly to the VBUS net from the USB-C connector. Place a 1\,\mu\text{F} (or higher) ceramic capacitor to PGND close to the pin.
* BATT / SYS:
  * BATT: Connect directly to the positive terminal of your 1-cell Li-Ion battery.
  * SYS: Connect to your system’s main input power rail to power the board downstream.
* GND / PGND: Tie to the ground plane with low-impedance vias.
* ISET / Configuration Resistors: Place the designated programming resistors from ISET to ground to configure your fast-charge current limit.
* Status Pins (STAT, INOKB): Pull up to a logic rail via 10\,\text{k}\Omega resistors if you want to monitor power-ok and charging states with your processor.
* ENBST: Tie low or control via logic if you intend to use the reverse-boost (OTG) 5V output capability on VBUS.

Here is the comprehensive, pin-by-pin connection guide for every IC in your block diagram, structured precisely for your OrCAD Capture schematic entry.

1. USB Type-C Receptacle (24-Pin Full-Featured)

* VBUS (Pins A4, A9, B4, B9): Connect together to the MAX77751 CHGIN input net and input bulk capacitors.
* GND (Pins A1, A12, B1, B12): Connect together to system Ground (GND/PGND).
* CC1 (Pin A5) & CC2 (Pin B5): Connect to TUSB322 pins CC1 and CC2.
* SuperSpeed TX/RX Lanes:
  * TX1+, TX1- (Pins A2, A3) & TX2+, TX2- (Pins B2, B3)
  * RX1+, RX1- (Pins B11, B10) & RX2+, RX2- (Pins A11, A10)
  * Route these 8 pins directly to the connector-side ports (C0p/C0n, C1p/C1n) of the HD3SS3212 mux.
* USB 2.0 Data Lines (D+/D-):
  * D+ (Pins A6, B6) & D- (Pins A7, B7)
  * Connect to the common ports (COM1/COM2) of the TS5USBA224 audio/USB switch.

* Sideband Use (SBU1, SBU2 - Pins A8, B8): Leave unconnected (unless implementing specific DisplayPort Alternate Mode, which is not required for standard USB data/audio/charging).
* Shield/Shell: Connect to chassis ground via an EMI suppression network (parallel 1\,\text{M}\Omega resistor and 10\,\text{nF} capacitor to GND).

2. TUSB322 (CC Controller — 12-Pin X2QFN)

* Pin 1 (CC1): Connect to USB-C Receptacle CC1.
* Pin 2 (CC2): Connect to USB-C Receptacle CC2.
* Pin 3 (VBUS_DET): Connect via a 900\,\text{k}\Omega external resistor to system VBUS.
* Pin 4 (DIR): Connect directly to the SEL pin of the HD3SS3212 mux to control lane orientation.
* Pin 5 (ADDR): Leave NC (No Connect) or tie according to the truth table to select GPIO mode.
* Pin 6 (INT_N / OUT3): Leave unconnected (or route to processor if interrupt monitoring is desired).
* Power & Ground Pins:
  * VDD: Connect to a stable 3.3V rail. Place a 0.1\,\mu\text{F} decoupling capacitor directly at the pin.
  * GND: Connect to system ground.

3. HD3SS3212 (2:1 SuperSpeed Mux — 20-Pin VQFN)

* Processor-Side Port (Port A — Pins 6, 7, 8, 9):
  * A0p, A0n, A1p, A1n: Connect through 100\,\text{nF} AC-coupling capacitors to your Main Processor's USB 3.0 TX and RX differential pairs.
* Connector-Side Ports (Ports B & C — Pins 12 through 19):
  * C0p, C0n, C1p, C1n (or B-side equivalents): Connect directly to the TX/RX differential pairs coming from the USB-C receptacle.
* Control & Power Pins:
  * SEL (Pin 5 or equivalent): Connect directly to the DIR output pin of the TUSB322.
  * OEn (Output Enable, Active-Low): Tie directly to GND to permanently enable the switch.
  * VCC: Connect to 3.3V rail with a local 0.1\,\mu\text{F} ceramic decoupling capacitor.
  * GND & Exposed Pad (EP): Solder firmly to the ground plane for thermal and

electrical integrity.

4. TS5USBA224 (USB 2.0 & Analog Audio Switch — 10-Pin UQFN)

 * Common Ports (COM1, COM2): Connect to the D+ and D- data lines coming from the USB-C connector.
 * Data Path (USB 2.0): Connect the USB switch outputs to the Main Processor USB 2.0 transceiver pins (USB_D+, USB_D-).
 * Audio Path (Analog Audio): Connect the audio switch outputs to the Audio Codec / DAC analog lines (HP_R, HP_L).
 * Control Pins (ASEL / Select / Enable): Driven by a processor GPIO or dedicated audio-jack detect line to toggle between USB Data Mode and Headset Audio Accessory Mode.
 * Power & Ground Pins:
   * VCC / VAUDIO: Connect to 3.3V supply rail with a local 0.1\,\mu\text{F} decoupling capacitor.
   * GND: Connect to system ground.

5. MAX77751 (Standalone USB-C Charger — 24-Pin FC2QFN)

 * CHGIN (Input Power): Connect directly to the VBUS net from the USB-C connector. Place a 1\,\mu\text{F} (or higher) ceramic capacitor from CHGIN to PGND close to the pin.
 * SYS (System Power Rail): Connect to your board's main input power rail. Place local bypass ceramic capacitors (e.g., 10\,\mu\text{F}).
 * BATT (Battery Terminal): Connect directly to the positive terminal of your 1-cell Li-Ion battery.
 * PGND / GND: Connect to power and signal grounds with low-impedance vias.
 * Configuration Pins:
   * ISET: Place a precise programming resistor from ISET to ground to set your fast-charge current limit.
   * ITOPOFF: Place a resistor to configure the top-off threshold (or ground/pull-up based on target configuration).
 * Status & Control Pins:
   * STAT: Pull up to a logic rail via a 10\,\text{k}\Omega resistor to monitor charging status via a microcontroller.
   * INOKB: Pull up to a logic rail via a 10\,\text{k}\Omega resistor to monitor input power-OK status.

* ENBST: Tie low (GND) to disable reverse-boost by default, or connect to a processor GPIO if you plan to use OTG 5V power output on VBUS.
* Switching Node (LX) / BYP: Connect an external 0.47\,\mu\text{H} power inductor between the LX pin and the internal switching node per the datasheet layout guidelines.

To make this as easy as possible for you to enter into OrCAD Capture, let’s break down your block diagram into a step-by-step wiring guide. Every single pin you need to connect is listed below, grouping them chip by chip.
(Note: In OrCAD Capture, you will place these parts as schematic symbols and use "nets" or wires with matching labels to connect them together.)
Step 1: The USB-C Receptacle (The Core Port)
The USB-C connector has 24 pins. Because it is reversible, many of the pins on top and bottom are mirrored. Here is how every functional group connects:
* VBUS Power Pins (Pins A4, A9, B4, B9):
  * Wire all four pins together into a single net named VBUS.
  * Connect this VBUS net directly to the input of your MAX77751 battery charger.
* Ground Pins (Pins A1, A12, B1, B12):
  * Wire all four pins together into a net named GND and connect them to your main ground plane.
* Configuration Channel Pins (CC1 & CC2):
  * Connect Receptacle Pin A5 (CC1) to TUSB322 Pin 1 (CC1).
  * Connect Receptacle Pin B5 (CC2) to TUSB322 Pin 2 (CC2).

* SuperSpeed High-Speed Data Lines (USB 3.0):
  * Receptacle pins for TX/RX (TX1+, TX1-, TX2+, TX2-, RX1+, RX1-, RX2+, RX2-) must be wired directly to the connector-side ports (C0p, C0n, C1p, C1n, etc.) of the HD3SS3212 Mux.
* USB 2.0 Data Lines (D+ & D-):
  * Connect Receptacle Pins A6/B6 (D+) together to COM1 on the TS5USBA224 switch.
  * Connect Receptacle Pins A7/B7 (D-) together to COM2 on the TS5USBA224 switch.
* Sideband Pins (SBU1, SBU2 - Pins A8, B8):
  * Leave these pins Unconnected (NC) since you are only doing standard data, audio, and charging.

Step 2: TUSB322 (CC Controller)

This chip figures out which way the cable is plugged in.

* Pin 1 (CC1): Wire to USB-C Receptacle Pin A5 (CC1).
* Pin 2 (CC2): Wire to USB-C Receptacle Pin B5 (CC2).
* Pin 3 (VBUS_DET): Connect via a 900\,\text{k}\Omega resistor to your VBUS net.
* Pin 4 (DIR): Wire this pin to a net named ORIENTATION_DIR. (This will tell your Mux how the cable is flipped).
* Pin 5 (ADDR): Leave Unconnected (NC).
* Pin 6 (INT_N): Leave Unconnected (NC).
* Power & Ground:
  * VDD Pin: Connect to your 3.3V power rail. Place a 0.1\,\mu\text{F} capacitor right next to this pin tied to ground.
  * GND Pin: Connect to GND.

Step 3: HD3SS3212 (SuperSpeed USB 3.0 Mux)

This chip switches your high-speed USB 3.0 data lines depending on how the cable is flipped.

* Connector-Side Differential Pairs:
  * Wire C0p, C0n, C1p, C1n directly to the TX/RX lines coming from your USB-C connector.
* Processor-Side Differential Pairs:
  * Wire A0p, A0n, A1p, A1n through 100\,\text{nF} capacitors straight to your Main Processor’s USB 3.0 TX and RX pins.
* Control & Power Pins:

* SEL Pin: Connect to the ORIENTATION_DIR net coming from the TUSB322 chip (Pin 4).
* OEn (Output Enable) Pin: Connect directly to GND (this keeps the switch permanently turned on).
* VCC Pin: Connect to your 3.3V power rail (with a local 0.1\,\mu\text{F} capacitor to GND).
* GND & Exposed Thermal Pad (EP): Connect firmly to GND.

Step 4: TS5USBA224 (USB 2.0 & Audio Switch)

This chip acts like a railway switch, sending D+/D- either to your processor (for regular USB data) or to your Audio Codec (for headphones).

* Common Pins (COM1 & COM2):
  * Connect to the D+ and D- lines coming straight from the USB-C connector.
* USB 2.0 Data Path:
  * Connect the data output pins to your Main Processor's USB 2.0 pins (USB_D+, USB_D-).
* Audio Path:
  * Connect the audio output pins to your Audio Codec / DAC audio lines.
* Select / Control Pin:
  * Connect to a GPIO pin from your main processor so it can tell the switch whether a data cable or an audio accessory is plugged in.
* Power & Ground:
  * VCC Pin: Connect to your 3.3V power rail (with a local 0.1\,\mu\text{F} capacitor to GND).
  * GND Pin: Connect to GND.

Step 5: MAX77751 (Battery Charger)

This chip takes the 5V from VBUS to charge your battery and power your board safely.

* CHGIN Pin: Connect directly to the VBUS net from your USB-C connector. Place a 1\,\mu\text{F} (or higher) capacitor from this pin to Ground.
* SYS Pin: This is your main system power output. Connect it to power the rest of your board's 3.3V/5V regulators. Place bulk capacitors (like 10\,\mu\text{F}) here.
* BATT Pin: Connect directly to the positive (+) terminal of your 1-cell Li-Ion battery.
* LX Pin: Connect an external 0.47\,\mu\text{H} power inductor between this pin and your internal switching node (as specified in the MAX77751 datasheet).

* PGND & GND Pins: Connect securely to your Ground plane.
* ISET Pin: Place a precise programming resistor from this pin to ground to set how fast your battery charges.
* Status Pins (STAT & INOKB):
  * Pull both pins up to your 3.3V rail using two separate 10\,\text{k}\Omega resistors so your processor can read if power is good and if the battery is charging.
* ENBST Pin: Connect directly to GND (this disables reverse-power output on VBUS, which you don't need unless you are using OTG power-bank mode).